

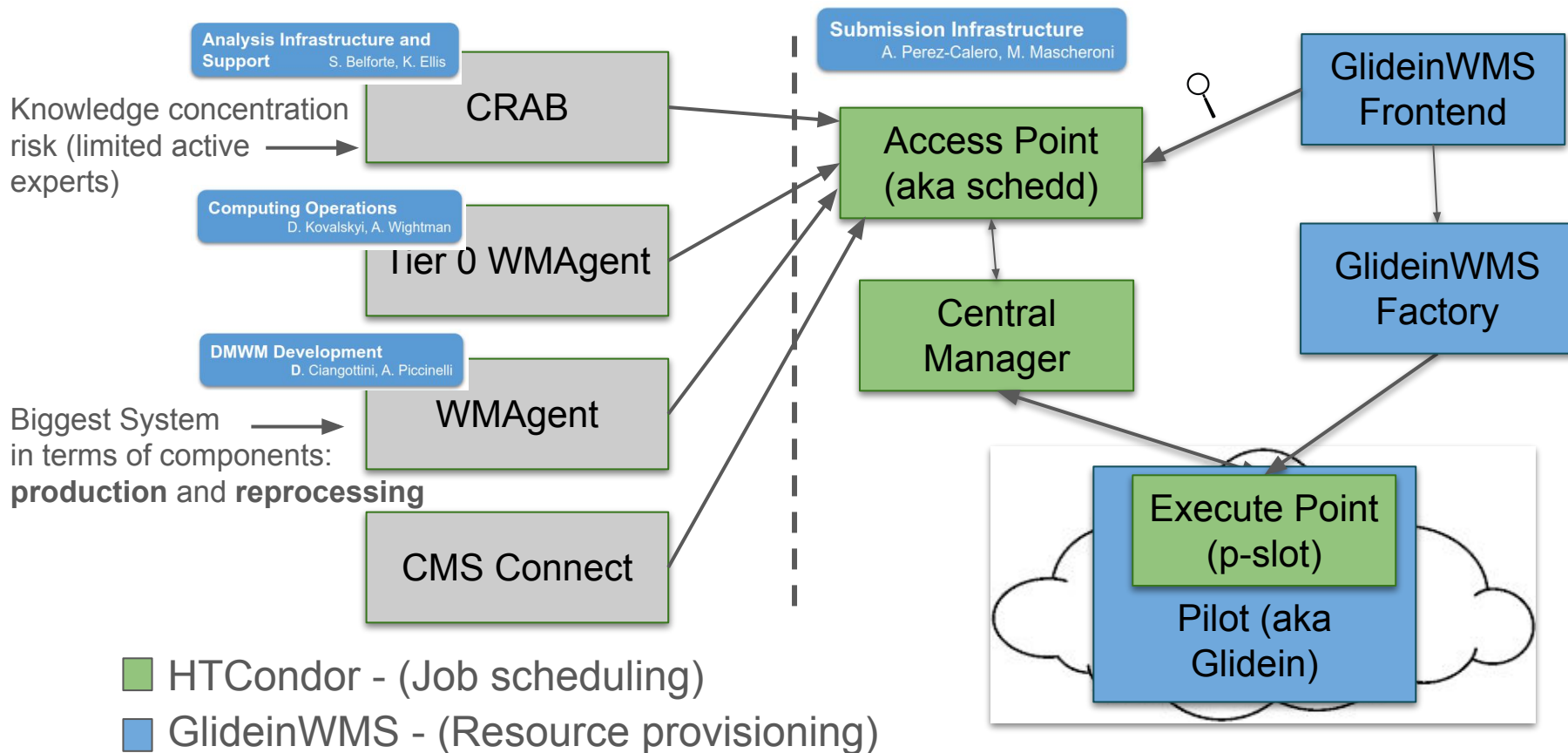
CMS O&C Week Summary: Workflow Management

Implications for GlideinWMS and Resource Provisioning
Marco Mascheroni

Submission Infrastructure in WM context

Workflows

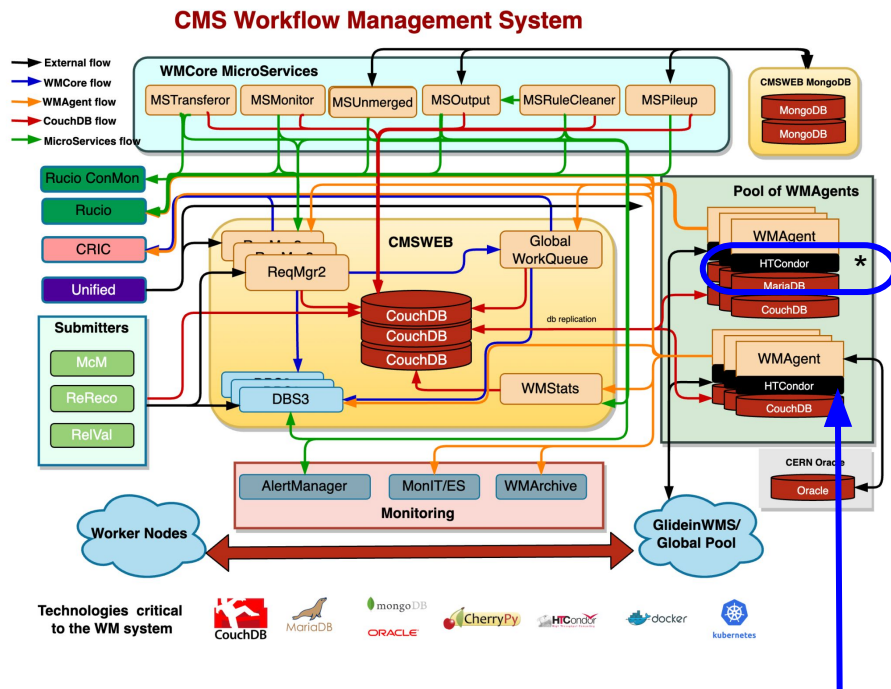
Jobs (SI)



Production and Reprocessing system

- Infrastructure designed more than 15 years ago
 - Works well even at scale
- Grew in **complexity** over the years
- **Doubts about sustainability** emerged
 - Mostly about ability to evolve the system to cope with the HL challenges

A committee was charged to review the whole CMS workflow management. Work started in October 2024



WMAgent itself has 16 independent components

<https://cms-wmcore.docs.cern.ch/wmcore/WMAgent/>

Review findings

Key Finding

CMS computing has successfully supported the full physics program from the preparation phase through Run 3, enabled by innovative development and specialized capabilities. However, as functional and scale requirements have grown, workflow management has evolved reactively—driven by operational needs rather than a coordinated, sustainable plan. The committee finds that CMS does not currently have a workflow management system suitable for Run 4 and beyond, nor a clear plan to develop or adopt one. The existing system is assessed by its experts as unsustainable and not evolvable. A viable path forward requires CMS to commit and sustain a multi-year effort to build or integrate a new system.

Committee results: <https://indico.cern.ch/event/1567220/>

Recommendation Step 0: Requirements and Planning

- Define requirements
- Evaluate external vs CMS solutions
- Build consensus

DMWM Workshop held in September 2025:

<https://indico.cern.ch/event/1554074/>

Recent progresses

- **Options evaluated** since September
 - CMS Revamp
 - PanDA
 - DIRAC
- WM Team members (Kenyi, Todor, Valentin, Alan) dedicated lots of effort in study of ATLAS and LHCb solutions
 - Meetings with ATLAS/LHCb experts
 - Deployments + PoCs
- Effort also in CMS led solution
 - Talked with HTCondor experts on a weekly base (Me, Andrea, Dima, Brian, Miron)
 - Developed two proof of concept based on **Microagent** idea
- **Last week's O&C week was a key turning point.**

08:30	12:30	Workflow Management Investigation Reports	40/S2-B01 - Salle Bohr	Join	40/S2-B01
Workflow Management Investigation Reports					
Conveners: Andrea Piccinelli (University of Notre Dame (US)), Diego Ciangottini (INFN, Perugia (IT))					
08:30		Introduction and summary from Fall O&C discussions	30m		
Speaker: Andrea Piccinelli (University of Notre Dame (US))					
WM day introduct... Zrec201.mp4					
09:00		System Architecture: PanDA and DIRAC integrations	30m		
Speakers: Alan Malta Rodrigues (University of Notre Dame (US)), Kevin Patrick Lannon (University of Notre Dame (US))					
WMS_SystemArc...					
09:30		System Architecture: in-house CMS HL-LHC solution	30m		
Speakers: Andrea Piccinelli (University of Notre Dame (US)), Marco Mascheroni (Univ. of California San Diego (US))					
In-house solution ...					
10:00		System Architecture: PoC of a micro-agent prototype	20m		
Speaker: Marco Mascheroni (Univ. of California San Diego (US))					
PoC of a micro-ag...					
10:20		Coffee break	20m		
10:40		System architecture: PoC for a light WMS prototype built on top of HTCondor DAGMan	30m		
A new light WMS prototype built on top of HTCondor					
Speaker: Dmytro Kovalskyi (Massachusetts Inst. of Technology (US))					
2026-03-17 Cond... Zrec202.mp4					
11:10		Supported workflows: Perspectives for Private Analyses Support (formerly, CRAB)	15m		
Speaker: Vijay Chakravarty (CERN)					
WMS_Compariso...					
11:25		Supported workflows: ATLAS and LHCb T0 operations	15m		
Speaker: Alan Malta Rodrigues (University of Notre Dame (US))					
WMS_SuportedW...					
11:40		Supported workflows: PanDA workflows and experience with CMS payload	20m		
Speaker: Alan Malta Rodrigues (University of Notre Dame (US))					
WMS_SuportedW...					
12:00		Supported workflows: DIRAC workflows and experience with CMS payload	20m		
Speaker: Todor Trendafilov Ivanov (University of Notre Dame (US))					
cmsdiracaux_pre... cmsdiracaux_pre... Zrec203.mp4					

13:30		Resource provisioning: Proposals for integrating GlideinWMS with DIRAC and PanDA	30m		
Speaker: Kenyi Paolo Hurtado Anampa (University of Notre Dame (US))					
Integrating Glidel...					
14:00		Design principles for integrating external workflow systems with the CMS Global Pool	30m		
Speaker: Marco Mascheroni (Univ. of California San Diego (US))					
Design principles ...					
14:30		Technology dependencies and code stack: insights for PanDA and DIRAC	30m		
Speaker: Valentin Y Kuznetsov (Cornell University (US))					
VK_techreview.pdf					
15:00		Deployment: Experience with deploying DIRAC	15m		
Speaker: Valentin Y Kuznetsov (Cornell University (US))					
15:15		Deployment: Experience with deploying PanDA	15m		
Speaker: Alan Malta Rodrigues (University of Notre Dame (US))					
WMS_Deploymen...					
15:30		Coffee break	20m		
15:50		Data Management integration in PanDA and DIRAC	40m		
Speaker: Diego Ciangottini (INFN, Perugia (IT))					
CMS Data Manag... Eric's report on Q...					
16:30		How to interact with DBS	20m		
Speaker: Todor Trendafilov Ivanov (University of Notre Dame (US))					

<https://indico.cern.ch/event/1612910/>

Tuesday dedicated to reports presentation

Wednesday: Workflow Management Development Contributions and Round Table

Round Table called by the CB Subgroup conveners (Oli and Markus) to:

- **Identify institutes and personpower** interested in developing and operating future CMS WMS
- Decide or at least discuss the **direction forward**

Decided to stop Panda investigation and focus on Dirac-X

Next steps:

- **Prepare requirements document based on September workshop and recent discussions**
- **Discuss potential collaboration with Dirac-X keeping a path for a CMS option open.**

This directly impacts how glideinWMS will integrate with future WM

What this means for glideinWMS

- glideinWMS remains central for resource provisioning
- Need to support:
 - DIRAC-X integration
 - possibly hybrid models
- Opportunity to:
 - simplify interfaces
 - modernize deployment
 - evaluate whether the micro-agent approach (from Brian B.) can be integrated into glideinWMS
 - [Architecture of CMS Led Solution](#)
 - [PoC of a micro-agent prototype](#)
 - [Condora WMS](#)